Higher National Diploma in Engineering (Mechanical/Electrical)
Forth Year, First semester Examination - 2020
ME 3227 – Industrial Engineering Management

Instructions for Candidates:
Answer any 5 questions.

No. of questions : 06
No. of pages : 03
Time : 03 hours

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01)

a) For which purpose OR is used? (02 marks)

b) Gama Safety Glove Company makes stock of two different types of gloves; a full fingered and a half fingered. The company has 960 hours in the cutting and sewing department, 800 hours in the finishing department and 120 hours in the packaging and shipping department, available per each day. The production time requirement and the profit per box of one dozen (12 units) are given bellow.

Type of glove	Production time (hours per box)			Profit per box (Rs.)
	Cutting & Sewing department	Finishing department	Packaging and Shipping department	
Full fingered	6	8	1	60.00
Half fingered	8	4	1	50.00

Assuming that the company wants to maximize profit,

i) Formulate LP model. (06 marks)
ii) By using graphical method find the number of boxes of each product that the company should manufacture per day in order to maximize the profit. (10 marks)
iii) What is the maximum profit that the company can achieve? (02 marks)

02)

A) a) Shortly describe the followings; (06 marks)

i) Comparison between Direct costs and Indirect costs

ii) Comparison between Variable costs and Semi variable costs

b) The cost of production of an ornament is as follows;

Fixed cost = Rs. 40,000.00 and Variable Cost = Rs. 50.00

If the selling price of one unit is Rs. 100.00, find the number of units to be produced in order to meet the break - even point. (04 marks)

B) a) Define the followings; (04 marks)

i) Utility ii) Demand

b) Discuss the advantages and disadvantages of oligopoly market structure in the point of view of a Sri Lankan consumer with the current market situation. (06 marks)

03)

a) Briefly describe followings; (04 marks)

i) IRR ii) Profitability Index

b) A manufacturing company is planning to purchase a new production machine and quotations have been called from two providers **A** & **B**.

- **A** offers the proposal for the machine at an initial cost of Rs.150,000/= at the start of 1st year and Rs.80,000/= equal installments at the start of next 5 years (from 2nd to 6th years) including periodical services.
- Proposal of **B** is to have an initial cost of Rs.200,000/= at the start of 1st year and Rs.100,000/= equal installments at the start of next 3 years (from 2nd to 4th years) with a separate Rs.10,000/= service charge at the start of next 5 years (from 2nd to 6th years).
- The annual income from machine **A** and **B** for the 1st year are Rs.120,000/= and Rs.110,000/= respectively. The income increases by 10% each and every year with respect to the previous year during next 6 years (from 2nd to 7th years). The salvage value of both machines are Rs.50,000/= at the end of this period.

i) Calculate the yearly Net Cash Flows of the two alternatives separately. (06 marks)

ii) Calculate the NPV for purchasing of both machines at a 10% discount rate. (08 marks)

iii) If you are the manager of manufacturing department, which alternative you are going to select and why? (02 marks)

04)

a) What are the goals of project management and planning? (04 marks)

b) Describe any 03 activities comes under the project life cycle. (06 marks)

c) Name any 4 types of organizational structures. (04 marks)

d) Describe the importance of building up an appropriate type of organizational structure for the successful operation of any company. (06 marks)

05)

a) What is the meaning of quality? (02 marks)

b) Distinguish the difference between quality control and quality assurance. (06 marks)

c) Name any 04 key functions of the human resource management. (04 marks)

d) Name any 04 characteristics of a good leader and describe the importance of them for a successful organization briefly. (08 marks)

06)

a) Give 04 advantages of facility layout design. (04 marks)

b) An assembly consists of the following elements as given in the table below. The production time available per day is 18 hours and the output is 72 units.

Task	A	B	C	D	E	F	G	H	I	J	K	L
Predecessor	-	A	B	B	B	B	C,D	G	E	I,F	H,J	K
Task time (min)	12	6	6	2	2	12	7	5	1	4	6	7

i) Draw the network diagram. (06 marks)

ii) Compute the cycle time per workstation in minutes. (02 marks)

iii) Compute the minimum number of workstations. (02 marks)

iv) How would you combine the tasks into workstations to minimize idle time? (04 marks)

v) What is the efficiency of line balancing? (02 marks)